Practical - 5

References, Citations, Headers and Footers

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Chapter 1

LaTeX Code

This chapter contains the complete LaTeX source code used to create this practical document.

1.1 Complete Source Code

The following code is the complete source code used to create this practical:

```
1
2  \documentclass[12pt,a4paper]{report}
3
4  % ---------- Packages ----------
5  \usepackage[utf8]{inputenc}
6  \usepackage[T1]{fontenc}
7  \usepackage{lmodern}
8
9  % Page layout
10 \usepackage{geometry}
11 \geometry{
12     a4paper,
13     margin=1in
14 }
15
16 % Images
17 \usepackage{graphicx}
18 \usepackage{float}
19
20 % Header and Footer
21 \usepackage{fancyhdr}
22
23 % Better formatting
24 \usepackage{setspace}
25 \usepackage{hyperref}
26
27 % Code listing
28 \usepackage{listings}
29 \usepackage{xcolor}
30
31 % Bibliography
32 \usepackage[
33     backend=biber,
34     style=ieee
35 ]{biblatex}
36
37 \addbibresource{references.bib}
38
39
40 % ---------- Code Listing Style ----------
41 \definecolor{codegreen}{rgb}{0,0.5,0}
42 \definecolor{codegray}{rgb}{0.5,0.5,0.5}
43 \definecolor{codepurple}{rgb}{0.58,0,0.82}
```

```
\definecolor{backcolour}{rgb}{0.95,0.95,0.95}

\lstdefinestyle{latexstyle}{
    backgroundcolor=\color{backcolour},
    commentstyle=\color{codegreen},
    keywordstyle=\color{blue},
    numberstyle=\tiny\color{codegray},
    stringstyle=\color{codepurple},
    basicstyle=\ttfamily\scriptsize,
    breakatwhitespace=false,
    breaklines=true,
    captionpos=b,
    keepspaces=true,
    numbers=left,
    numbersep=8pt,
    showspaces=false,
    showstringspaces=false,
    showtabs=false,
    frame=single,
    tabsize=4
}

\lstset{style=latexstyle}


% ---------- Header/Footer ----------
\pagestyle{fancy}
\fancyhf{}

\fancyhead[L]{References and Citations}
\fancyhead[R]{Overleaf Practical}

\fancyfoot[L]{Department of Computer Science}
\fancyfoot[R]{Page \thepage}

\renewcommand{\headrulewidth}{0.4pt}
\renewcommand{\footrulewidth}{0.4pt}


% ---------- Document ----------
\begin{document}



% TITLE PAGE


\begin{titlepage}
    \centering

    \vspace*{2cm}

    {\Huge\bfseries Practical - 5\par}

    \vspace{1.5cm}

    {\Large References, Citations, Headers and Footers\par}

    \vspace{2cm}

    \vfill

\end{titlepage}



% ROMAN PAGE NUMBERING


\pagenumbering{roman}

% Table of Contents
\tableofcontents
```

```
\clearpage



% ARABIC PAGE NUMBERING


\pagenumbering{arabic}
\setcounter{page}{1}




\chapter{LaTeX Code}

This chapter contains the complete LaTeX source code used
to create this practical document.

The \texttt{listings} package is used to display the source
code in a formatted and readable manner. Line numbers,
syntax highlighting, and a frame are included to make the
code easier to understand.

\section{Complete Source Code}

The following code is the complete source code used to
create this practical:

\begin{lstlisting}[language={[LaTeX]TeX}]

% The complete source code is displayed in this chapter.


\chapter{References and Citations}

\section{Adding a Citation}

A citation can be inserted using the
\texttt{\textbackslash cite} command.

For example:

\begin{verbatim}
LaTeX is widely used for technical documents \cite{lamport1994}.
\end{verbatim}

This produces a numbered citation such as [1].

Another source can be cited using:

\begin{verbatim}
Computer networks are discussed in \cite{kurose2017}.
\end{verbatim}

Computer networks are an important area of computer science
\cite{kurose2017}.


\section{Multiple Citations}

Multiple references can be cited together:

\begin{verbatim}
\cite{lamport1994,kurose2017}
\end{verbatim}

For example, several important concepts in computer science
are discussed in the literature
\cite{lamport1994,kurose2017}.



```

```latex

\chapter{Headers and Footers}

Headers and footers provide information at the top and
bottom of every page.

The \texttt{fancyhdr} package is used to create customized
headers and footers.

The header contains:

\begin{itemize}
    \item Document title
    \item Practical title
\end{itemize}

The footer contains:

\begin{itemize}
    \item Department name
    \item Current page number
\end{itemize}

The page number is automatically generated using:

\begin{verbatim}
\thepage
\end{verbatim}



\chapter{Page Numbering}

Different page numbering styles can be used for different
parts of a document.

\section{Roman Numerals}

The preliminary pages use Roman numerals.

This is enabled using:

\begin{verbatim}
\pagenumbering{roman}
\end{verbatim}

The pages are numbered:

\begin{center}
i, ii, iii, iv, ...
\end{center}


\section{Arabic Numerals}

The main content uses Arabic numerals.

This is enabled using:

\begin{verbatim}
\pagenumbering{arabic}
\setcounter{page}{1}
\end{verbatim}

The pages are numbered:

\begin{center}
1, 2, 3, 4, ...
\end{center}



\chapter{Table of Contents}
```

```
263
264 LaTeX can automatically generate a Table of Contents.
265
266 The command used is:
267
268 \begin{verbatim}
269 \tableofcontents
270 \end{verbatim}
271
272 Chapter and section headings are automatically added to
273 the Table of Contents.
274
275 For example:
276
277 \section{Automatic Updating}
278
279 When chapters or sections are added, removed, or renamed,
280 LaTeX automatically updates the Table of Contents after
281 recompiling the document.
```

Chapter 2

References and Citations

2.1 Adding a Citation

A citation can be inserted using the `\cite` command.
For example:

```
LaTeX is widely used for technical documents \cite{lamport1994}.
```

This produces a numbered citation such as [1].
Another source can be cited using:

```
Computer networks are discussed in \cite{kurose2017}.
```

Computer networks are an important area of computer science [1].

2.2 Multiple Citations

Multiple references can be cited together:

```
\cite{lamport1994,kurose2017}
```

For example, several important concepts in computer science are discussed in the literature [1], [2].

Chapter 3

Headers and Footers

Headers and footers provide information at the top and bottom of every page.

The `fancyhdr` package is used to create customized headers and footers.

The header contains:

- Document title
- Practical title

The footer contains:

- Department name
- Current page number

The page number is automatically generated using:

```
\thepage
```

Chapter 4

Page Numbering

Different page numbering styles can be used for different parts of a document.

4.1 Roman Numerals

The preliminary pages use Roman numerals.
This is enabled using:

```
\pagenumbering{roman}
```

The pages are numbered:

i, ii, iii, iv, ...

4.2 Arabic Numerals

The main content uses Arabic numerals.
This is enabled using:

```
\pagenumbering{arabic}
\setcounter{page}{1}
```

The pages are numbered:

1, 2, 3, 4, ...

Chapter 5

Table of Contents

LaTeX can automatically generate a Table of Contents.
The command used is:

```
\tableofcontents
```

Chapter and section headings are automatically added to the Table of Contents.
For example:

5.1 Automatic Updating

When chapters or sections are added, removed, or renamed, LaTeX automatically updates the Table of Contents after recompiling the document.

References

[1] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Pearson, 2017.

[2] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.